

QIHANG LI

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RESEARCH INTERESTS

My research focuses on robot collaboration: enabling robots to work with teammates, both other robots and humans, through perception, task allocation, and coordinated task execution. I draw on large-scale pre-trained models, learning-based perception, and planning to operate under uncertainty and limited communication, and I am passionate about making my algorithms run reliably and efficiently on real robots.

EDUCATION

North Carolina State University, US <i>PhD in Computer Science. Advisor: Dr. Peng Gao</i>	Fall 2025 – Present GPA: 4.00
Johns Hopkins University, US <i>MSE in Robotics</i>	Sep 2021 – May 2023 GPA: 3.86
University of Liverpool, UK	Sep 2019 – May 2021
Xi'an Jiaotong-Liverpool University, China <i>BEng in Mechatronics and Robotics System (Joint Degree)</i>	Sep 2017 – Sep 2019 First Class Honor Degree

RESEARCH EXPERIENCE

Research Associate in CoRob Lab , North Carolina State University <i>Advised by Dr. Peng Gao</i>	Jul 2025 – Present
• Collaborative Perception: Developing vision-based perception for robot teams, including communication-efficient relative pose estimation between robots under limited bandwidth and intermittent shared views. [website][paper] • Robot Collaboration: Developing task coordination for a hybrid robot team of tabletop manipulators, mobile robots, and mobile manipulators, building on preliminary work in long-horizon task and motion planning. • Real-Robot Deployment: Independently ran real-world experiments across multiple projects on a Unitree quadruped, manipulators, a mobile manipulator, and wheeled robots, and built the perception and navigation stacks.	
Research Associate in Mazurowski Lab , Duke University <i>Advised by Dr. Maciej A. Mazurowski</i>	Oct 2024 – Jun 2025
• Contributed to MRI-CORE, a large-scale pre-trained model for MRI that transfers to data-restricted segmentation, classification, and zero-shot segmentation tasks. [paper] • Contributed to multiple other projects, including volumetric image annotation with SAM 2, breast MRI registration (Best Paper Award, MICCAI DeepBreath Workshop 2025), and CT-based bone loss measurement. • Developed BreastSegNet, a multi-label segmentation model for breast MRI. [paper]	
Research Associate in SAIR Lab , University at Buffalo <i>Advised by Dr. Chen Wang and Dr. Junyi Geng</i>	Aug 2023 – Sep 2024
• Enhanced the Model Predictive Control (MPC) module in the open-source library PyPose, enabling robust operation and highly efficient differentiation. [website][paper] • Developed an end-to-end training pipeline that embeds MPC within the neural network, enabling explicit prediction of kinematically feasible trajectories for motion planning. [website][paper]	
Research Assistant in BIGSS Lab , Johns Hopkins University <i>Advised by Dr. Mehran Armand</i>	Sep 2022 – Jul 2023
• Developed RGB camera-based state-estimation tools for Continuum Dexterous Manipulators (CDMs). • Designed a novel capacitance-based sensor for CDM state estimation at 1% of the cost of the original solution, with sub-centimeter accuracy on a prototype ten times larger. [paper]	

PREPRINTS AND PUBLICATIONS

Bold titles indicate first-author papers; [†] indicates co-first authorship.

PREPRINTS

- [P4] **Skeleton-Conditioned Base Placement Evaluation for Long-Horizon Mobile Manipulation.** [\[paper\]](#)
- [P3] Execution-time Perceptual Degradation Diagnosis and Recovery for Resilient Robot Systems.
- [P2] Automated Glenoid Bone Loss Measurement and Segmentation in CT Scans for Pre-operative Planning in Shoulder Instability. [\[paper\]](#)
- [P1] MRI-CORE: A Foundation Model for Magnetic Resonance Imaging. [\[paper\]](#)

PUBLICATIONS

Robot Collaboration, Perception, and Planning

- [11] **Communication-Efficient Relative Pose Estimation with Vision Foundation Models for Ephemeral Collaborative Perception.** *International Conference on Intelligent Robots and Systems (IROS)*, 2026. [\[website\]](#) [\[paper\]](#)
- [10] **iKap: Kinematics-aware Planning with Imperative Learning.** *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [\[website\]](#) [\[paper\]](#)
- [9] Imperative Learning: A Self-supervised Neural-Symbolic Learning Framework for Robot Autonomy. *The International Journal of Robotics Research (IJRR)*, 2025. [\[paper\]](#)
- [8] PyPose v0.6: The Imperative Programming Interface for Robotics. *International Conference on Intelligent Robots and Systems (IROS) Workshop*, 2023. [\[paper\]](#)

Medical Robotics

- [7] A Hierarchical Laser Control Strategy for Spatially Uniform Thermal Dose Delivery in Laser-Induced Thermal Therapy. *IEEE/ASME Transactions on Mechatronics (TMeCh)*, 2025. [\[paper\]](#)
- [6] **Advancing Hand Rehabilitation: A Novel 3D-Printed Soft Pneumatic Glove for Enhanced Mixed Reality Interaction[†].** *IEEE Haptics Symposium*, WIP paper, 2024. [\[paper\]](#)
- [5] **Data-driven Shape Sensing of Continuum Dexterous Manipulators Using Embedded Capacitive Sensor.** *IEEE Sensors 2023 (Top 5% of the accepted papers)*. [\[paper\]](#)

Medical Image Analysis

- [4] GuidedMorph: Two-Stage Deformable Registration for Breast MRI. *IEEE Journal of Biomedical and Health Informatics (J-BHI)*, 2025. [\[paper\]](#)
- [3] Accelerating Volumetric Medical Image Annotation via Short-Long Memory SAM 2. *IEEE Transactions on Medical Imaging (TMI)*, 2025. [\[paper\]](#)
- [2] **BreastSegNet: Multi-label Segmentation of Breast MRI.** *MICCAI, DeepBreath Workshop*, 2025. [\[paper\]](#)
- [1] Are Vision Foundation Models Ready for Out-of-the-Box Medical Image Registration? *MICCAI, DeepBreath Workshop*, 2025 (**Best Paper Award**). [\[paper\]](#)

TEACHING

Teaching Assistant Statistical Learning for Engineers, JHU	2022 Fall
Teaching Assistant Robot Motion Planning, NCSU	2025 Fall

ACADEMIC SERVICES

- Reviewer** IEEE International Conference on Intelligent Robots and Systems (IROS)
- Reviewer** IEEE International Conference on Robotics and Automation (ICRA)
- Reviewer** IEEE Robotics and Automation Letters (RA-L)